

Distillation of GPT-2 Small

Compressing a 124M-parameter teacher into a 30M student on WikiText-2

Yicheng Yao · Jarvis Wang · Jiangchuan Yu

ECE1508 · Deep Generative Models · Summer 2026

01 · INTRODUCTION & MOTIVATION

WHAT IT IS

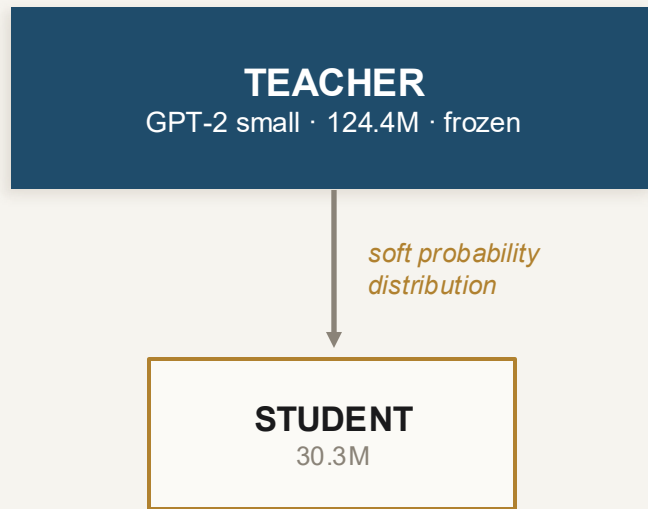
Knowledge distillation trains a small student to imitate a large teacher — learning from the teacher's full probability distribution, not only the single correct next token.

WHY IT MATTERS

- Lower latency
- Smaller size
- Cheaper deployment
- *Keep the quality while cutting the cost.*

RESEARCH QUESTIONS

- Does teacher guidance beat corpus-only?
- Is imitating the teacher alone enough?
- Which temperature transfers best?
- Quality vs. efficiency trade-off?



The teacher's soft distribution is the training signal.

02 · DESIGN STRUCTURE

Same architecture, same data, same loop — only the loss changes.

Teacher = frozen GPT-2 small, 12 layers / 768 hidden / 12 heads (124.4M) · Student = 6 layers / 384 hidden / 6 heads (30.3M)
· shared tokenizer, 50,257 vocab, 1024 context

Corpus only

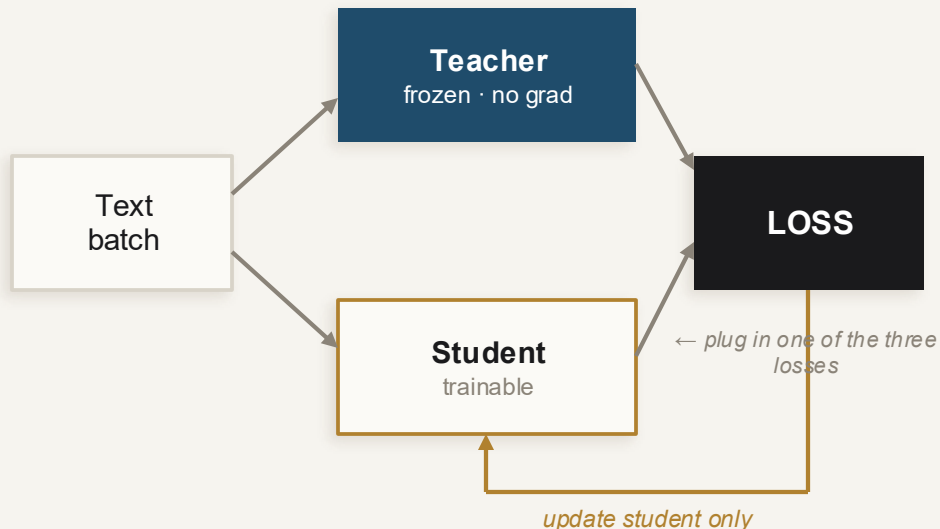
$$L = CE(student, true\ token)$$

Teacher only

$$L = KL(teacher \parallel student) \cdot T^2$$

Teacher + corpus

$$L = \alpha \cdot CE + (1-\alpha) \cdot KL \cdot T^2 \quad (\alpha = 0.5)$$



Fair-comparison controls: identical init weights · reseed per run (same data order & dropout) · early stopping with best-weight restore.

03 · TEMPERATURE SWEEPING

WHY

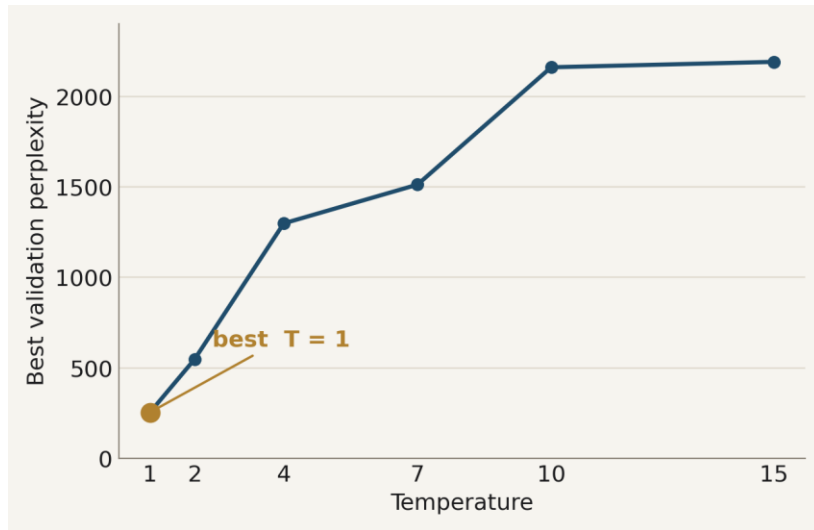
Temperature softens the teacher's distribution — too sharp hides the “dark knowledge,” too soft washes the signal out.

HOW

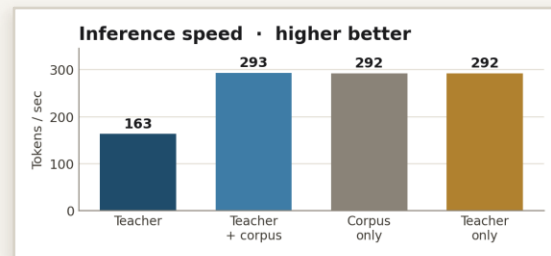
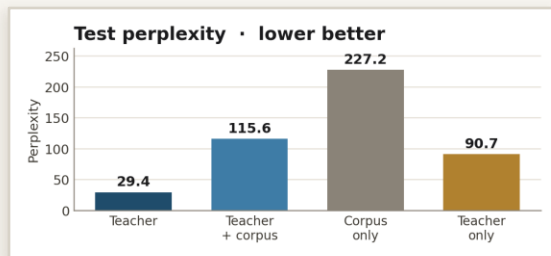
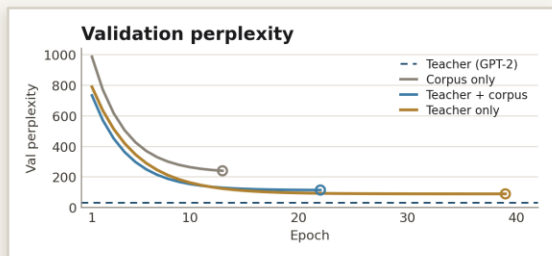
Train a student per T on the teacher-only objective, then rank by validation perplexity.

Best at $T = 1$. Perplexity explodes at high T ($\approx 2000+$ at $T = 10-15$).

Small student $\rightarrow$ focus on the top token $\rightarrow T = 1$ wins.



04 · FINAL RESULTS & MODEL COMPARISON



FINDINGS

- **Distillation clearly helps** — the teacher signal cuts perplexity $\sim 2\times$ vs corpus-only.
- **Surprise: teacher-only beat teacher+corpus** — on a small corpus the hard labels overfit (corpus-only stopped at epoch 13), while the soft target is a smoother, richer signal (teacher-only ran to epoch 39).
- **Still well behind the teacher (90.7 vs 29.4)** — GPT2-Small has never even trained on WikiText, so the true gap is larger.

EFFICIENCY

4.1× fewer parameters · 8× on the transformer stack

119 MB checkpoint · vs 475 MB teacher

1.8× faster generation vs the teacher

FUTURE WORK

Tune α (jointly with T) · more data (WikiText-103) / longer training · larger students.

Takeaway: distillation works — a $4\times$ smaller, $1.8\times$ faster student keeps usable language ability, and the teacher's soft targets were the key ingredient ($\sim 2\times$ better than corpus-only). On a small dataset, learning from the teacher alone worked best — soft targets act as a strong regularizer.